Ex.No.4 - Image Histogram and Histogram Equalization

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In [27]:
```
import numpy as np
import cv2
import matplotlib.pyplot as plt
```

In [28]:
```
def equalize_hist(image, L=256):
    # Step 1: Find unique intensity values and their counts
    intensity_levels, counts = np.unique(image, return_counts=True)
    total_pixels = image.size

    # Step 2: Calculate PDF and CDF
    pdf = counts / total_pixels
    cdf = np.cumsum(pdf)

    # Step 3: Create transformation function
    transform = np.floor((L - 1) * cdf).astype(np.uint8)

    # Step 4: Create a mapping dictionary and apply the mapping
    mapping = dict(zip(intensity_levels, transform))
    equalized_image = np.vectorize(mapping.get)(image)

    return equalized_image.astype(np.uint8)
```

In [29]:
```
# Read image in grayscale
path = r"D:\VI\IP & CV\Lab Experiments\Inputs\roses.jpg"
image = cv2.imread(path, cv2.IMREAD_GRAYSCALE)

# Custom histogram equalization
custom_image = equalize_hist(image)

# OpenCV histogram equalization
opencv_image = cv2.equalizeHist(image)
```

In [34]:
```
# Display images
plt.figure(figsize=(12,4))

plt.subplot(1,3,1)
plt.imshow(image, cmap='gray')
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(custom_image, cmap='gray')
```

```
plt.title("Custom Equalized Image")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(opencv_image, cmap='gray')
plt.title("OpenCV Equalized Image")
plt.axis("off")

plt.tight_layout()
plt.show()
```

In [35]:

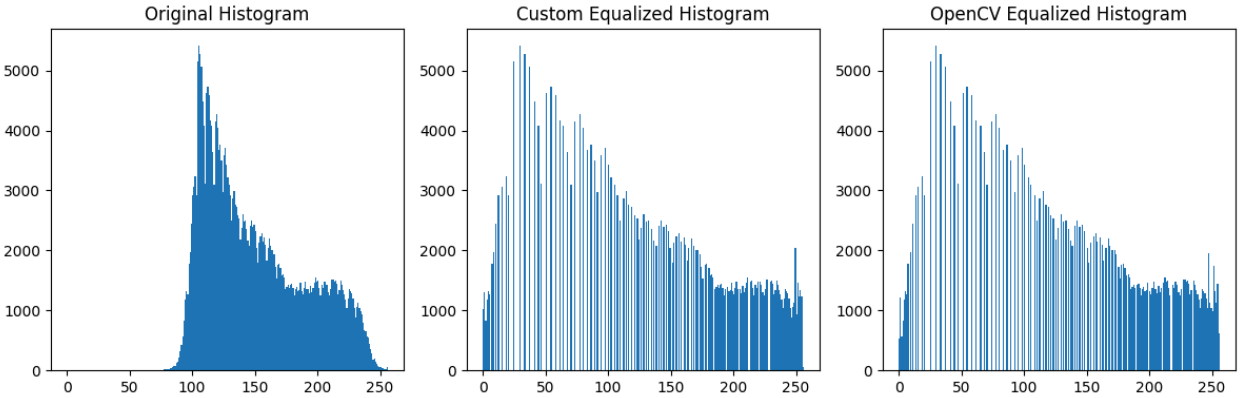
```
# Display Histograms
plt.figure(figsize=(12,4))

plt.subplot(1,3,1)
plt.hist(image.ravel(), bins=256, range=[0,256])
plt.title("Original Histogram")

plt.subplot(1,3,2)
plt.hist(custom_image.ravel(), bins=256, range=[0,256])
plt.title("Custom Equalized Histogram")

plt.subplot(1,3,3)
plt.hist(opencv_image.ravel(), bins=256, range=[0,256])
plt.title("OpenCV Equalized Histogram")

plt.tight_layout()
plt.show()
```



In [ ]: